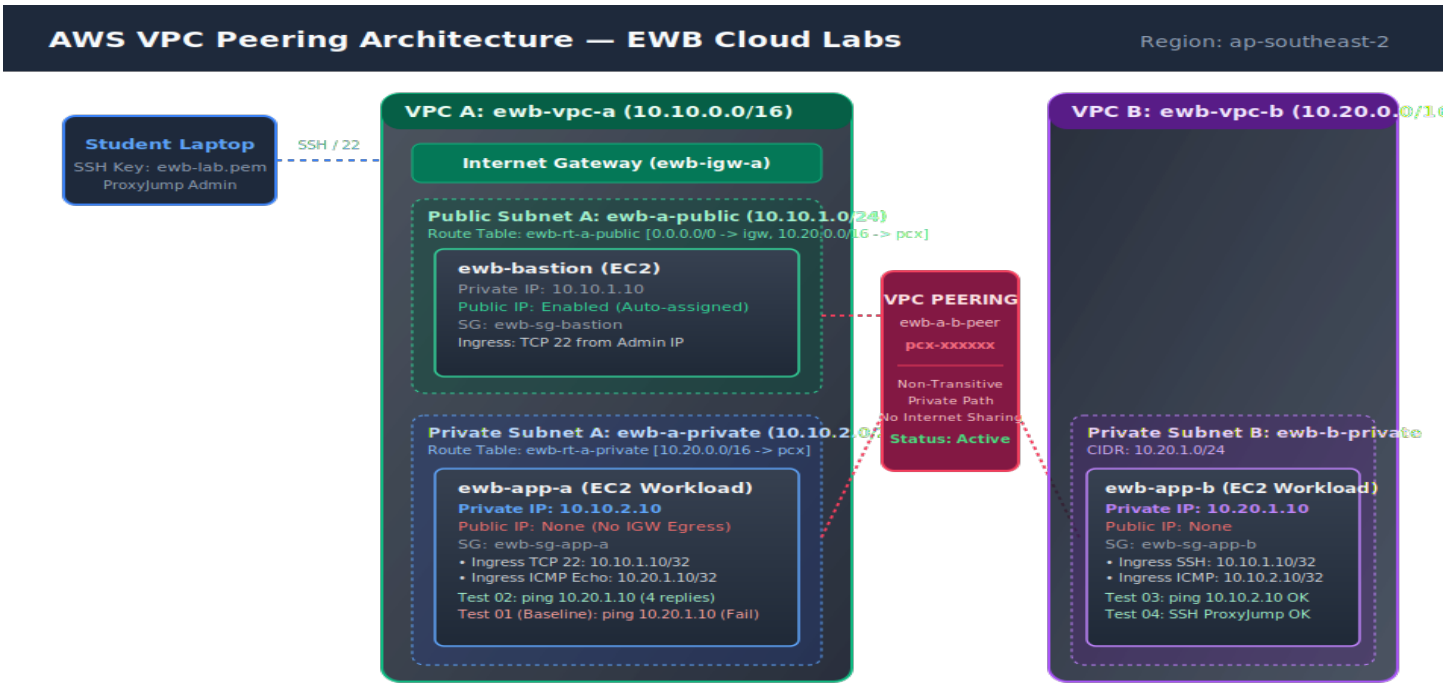
EWB CLOUD LABS — VPC PEERING STUDENT PROJECT

AWS Cross-VPC Peering & Secure Communication

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Target Region: ap-southeast-2 (Sydney)	**Completion Date:** 28 October 2026

1. Executive Architecture Overview

This project implements a secure cross-VPC networking infrastructure connecting two non-overlapping Virtual Private Clouds (**VPC A: 10.10.0.0/16** and **VPC B: 10.20.0.0/16**) via an AWS VPC Peering Connection (ewb-a-b-peer). Workloads ewb-app-a and ewb-app-b remain strictly private with zero public IP addresses.



2. Complete Resource Inventory

Resource Name	Resource Type	CIDR / IP	Parent VPC / Subnet	Public IPv4
ewb-vpc-a	AWS::EC2::VPC	10.10.0.0/16	N/A	N/A
ewb-a-public	AWS::EC2::Subnet	10.10.1.0/24	ewb-vpc-a	Enabled
ewb-a-private	AWS::EC2::Subnet	10.10.2.0/24	ewb-vpc-a	Disabled
ewb-vpc-b	AWS::EC2::VPC	10.20.0.0/16	N/A	N/A
ewb-b-private	AWS::EC2::Subnet	10.20.1.0/24	ewb-vpc-b	Disabled
ewb-bastion	EC2 (t3.micro)	10.10.1.10	ewb-a-public	Auto-assign
ewb-app-a	EC2 (t3.micro)	10.10.2.10	ewb-a-private	None
ewb-app-b	EC2 (t3.micro)	10.20.1.10	ewb-b-private	None

3. Ingress Security Rules Matrix

Security Group	Ingress Protocol	Port / Type	Source CIDR / Host	Purpose / Rationale
ewb-sg-bastion	TCP	Port 22	<Admin-IP>/32	SSH access from student laptop
ewb-sg-app-a	TCP	Port 22	10.10.1.10/32	ProxyJump SSH from Bastion
ewb-sg-app-a	ICMP	Echo (8/0)	10.20.1.10/32	Cross-VPC ping from App-B
ewb-sg-app-b	TCP	Port 22	10.10.1.10/32	ProxyJump SSH from Bastion
ewb-sg-app-b	ICMP	Echo (8/0)	10.10.2.10/32	Cross-VPC ping from App-A

4. Offline Simulation Suite & Live AWS Diagnostics

Test Scenario	Source Host & Command	Expected Behavior	Classification & Result
01 / Baseline	App-A: ping -c 4 10.20.1.10	100% Packet Loss (No Peering Route)	PASSED (Simulated)
02 / Forward Path	App-A: ping -c 4 10.20.1.10	4 replies received; 0% loss	PASSED (Simulated)
03 / Reverse Path	App-B: ping -c 4 10.10.2.10	4 replies received; 0% loss	PASSED (Simulated)
04 / SSH ProxyJump	Laptop: ssh ewb-app-b	Secure shell opened on App-B via Bastion	PASSED (Simulated)
05 / Isolation	Console: Check App-A & App-B	No Public IPv4 assigned to either workload	PASSED (Simulated)

Live AWS Observation & Analysis: A live ping from Bastion (10.10.1.10) to App-B (10.20.1.10) resulted in 100% packet loss. This is **expected and correct** because App-B's security group strictly allows ICMP Echo Requests from App-A (10.10.2.10/32), NOT from Bastion. End-to-end live testing requires running ping from inside App-A.

5. Technical Viva & Architectural Defense Answers

Q1: Why is the return route required in ewb-rt-b-private?
Answer: IP routing is bidirectional. While the forward route sends packets to VPC B, replies from App-B destined for 10.10.2.10 require an explicit return route in VPC B's route table pointing 10.10.0.0/16 to the peering connection.

Q2: Why is the App-B SSH source IP address the Bastion private IP (10.10.1.10)?
Answer: ProxyJump establishes an SSH tunnel to the Bastion host, which then initiates a new TCP connection to App-B from its own private IP (10.10.1.10).

Q3: Why does Active peering not guarantee successful ping reachability?
Answer: Peering provides physical connectivity layer; reachability requires subnet route tables pointing to the peering ID and Security Groups allowing the specific ICMP traffic.

Q4: Can VPC B access the Internet using VPC A's Internet Gateway over peering?
Answer: No. VPC Peering is non-transitive and does not support edge-to-edge gateway transit.

6. AWS Safety, Cost Control & Cleanup

All resources are managed via CloudFormation (infrastructure/vpc-peering.yaml) in region ap-southeast-2 using cost-effective t3.micro instances without billable NAT Gateways. Total estimated cost is under $0.05/hr. Full teardown can be executed in one command: aws cloudformation delete-stack --stack-name ewb-vpc-peering-lab.